

Course Outline for DevOps Training with Multi-Cloud and GenAI

Training Details:

Course Name	DevOps with MultiCloud and GenAI
Level	Beginner to Expert
Duration	130 hrs
Participant background (Prerequisites)	Basic Understanding of Cloud Concepts, Networking,DNS Concepts

Break-Down of Learning Paths:

Topics	Number of Days
Generic DevOps	54 (2 Hours/Day)
Azure Associate	25 (2 Hours/Day)
AWS Solution Architect	25 (2 Hours/Day)
GenAI Integration with DevOps	26 (2 Hours/Day)
Total	10 (4 Hours/Day)

Course Contents DevOps with MultiCloud & GenAI

Generic DevOps: 54 hrs

Module 1: DevOps Overview

S.No.	Content	hrs assigned
1	Evolution of Waterfall, Agile and DevOps, DevOps Benefits, Stages, Lifecycle, CI/CD, Automation in DevOps	2 hrs

Module 2: Fundamentals of Linux Operating System

S.No.	Content	hrs assigned
1	Overview of Linux, Linux Architecture, Linux Distributions, Basic Linux Commands	2 hrs
2	File Permission Management, User creation, SSH, Vi utility	2 hrs
3	Shell Scripting	2 hrs

Module 3: Fundamentals of Python scripting

S.No.	Content	hrs assigned
1	Overview,Features, Benefits, Uses of Python,Installation and Setup of Python Environment, Various Types of Sequences in Python	2 hrs
2	Loops and Control flow,File Operations,Python Functions,	2 hrs
3	OOPs Concepts,Modules,Errors and Exception Handling	2 hrs

Module 4: Version Control System using - Git and GitHub

S.No.	Content	hrs assigned
1	Overview of Version Control System,Central vs Distributed Version Control System, Installation of Git, Git File workflow,Important Git Commands, Creating and Managing git Repositories, GIT-IGNORE	2 hrs
2	GIT Misc Commands,Reverting and Resetting ,GIT Stash,GIT Branching Strategies, working with Git Branching, Merging, Rebase and Squash, Introduction to GitHub, Managing Remote Repositories,Handling Github repositories using Visual Studio Code	2 hrs

Module 5: Maven Build Tool

S.No.	Content	hrs assigned
1	Overview of Various Build Tools,Maven Architecture, Maven Plugins,Maven commands	2 hrs
2	Building of web applications, standalone applications, enterprise applications using Maven	2 hrs

Module 6: Continuous Integration Using Jenkins

S.No.	Content	hrs assigned
1	Difference between Continuous vs Traditional Integration, Overview of Jenkins, Jenkins Installation and Configuration, Jenkins Plugins, Jenkins Freestyle and Pipeline Jobs	2 hrs
2	Scripted and Declarative Pipelines,Jenkins Master-Slave Architecture, Configuring Slave Node to Jenkins	2 hrs
3	Configure Tomcat Server,Integrate and Deploy to Tomcat Server using Jenkins, Jenkins Build Triggers, Enable Email Notifications, Jenkins Management	2 hrs

Module 7: Containerization, Docker, and Docker Hub

S.No.	Content	hrs assigned
1	Introduction to Virtualization and Containerization, Docker Architecture, Overview of Docker Hub, Docker Installation, Docker Commands, Container Modes, Port Binding	2 hrs
2	Docker Volume, Docker Networking	2 hrs
3	Docker file, Customized Docker Images, Running and Managing Containers, Docker Compose, Overview of Docker Swarm Orchestration tool with all commands, Jenkins Integration with Docker	2 hrs

Module 8: Container Orchestration Tool – Kubernetes

S.No.	Content	hrs assigned
1	Overview of Container Orchestration, Difference between Docker swarm and Kubernetes Cluster, Kubernetes Architecture, Installation of Kubernetes – Minikube, Kops script, Managed AKS/EKS/GKE Cluster [anyone], Namespaces	2 hrs
2	Kubernetes Nodes, Kubernetes Pods, Kubernetes Deployments, Rolling updates and rollbacks, Scaling up and down of the application	2 hrs
3	Services in Kubernetes, Kubernetes Volume -empty dir, HostPath, Persistent Volume, Persistent Volume Claim, Dynamic Provisioning, Storage Classes, Stateful application deployment, Pod Scheduling, Node Scheduling	2 hrs

Module 9: Infrastructure as Code (IaC) using Terraform

S.No.	Content	hrs assigned
1	Introduction to Terraform, Terraform Vs Ansible, Terraform Architecture, Terraform Configuration HCL files	2 hrs
2	Terraform Commands, Managing Terraform Resources on AWS like ec-2 instance along with VPC, Subnet, Key-pair value, Security Group, Internet Gateway,EBS, Input, Output variables, Dynamically fetch AMI, Primitive and Complex Data types, Terraform functions.	2 hrs
3	Local Values, Data Sources, Remote State using AWS S3 Bucket, Local, Remote exec Provisioners, Terraform Modules	2 hrs

Module 10: Configuration Automation using Ansible

S.No.	Content	hrs assigned
1	Overview of Configuration Automation, Introduction to Ansible, Ansible Architecture, Components of Ansible, Installation and Configuration of Ansible, Ansible ad-hoc commands	2 hrs
2	Ansible Master and Slave Set-up, Ansible Playbooks	2 hrs
3	Ansible Variables, Ansible Handlers, Ansible Role using Ansible Galaxy	2 hrs

Module 11: Continuous Monitoring using Prometheus and Grafana

S.No.	Content	hrs assigned
1	Overview of continuous monitoring, Continuous monitoring tools in DevOps, Installation and Configuration of Prometheus and Grafana, Prometheus Architecture, Monitoring using Prometheus, Dashboard visualization using Grafana	2 hrs

Microsoft Azure Cloud – Associate: 25 hrs**Module 1: Getting Started with Microsoft Azure**

S.No.	Content	hrs assigned
1	Overview of cloud computing, Explore Microsoft Azure,IaaS and PaaS Services provided by Microsoft Azure, Demonstrating the working of the Azure Portal, Usage and understanding of Azure Command Line Interface (CLI), Working with the Azure PowerShell	2 hrs

Module 2: Exploring the Microsoft Entra ID

S.No.	Content	hrs assigned
1	Introduction to Microsoft Entra IID, Compare the difference between Windows AD and Azure AD, Identity and access management in Azure, Users,Group creation, Azure AD domains, Microsoft Entra tenant, Authentication options, Azure AD Connect, Self-service password reset (SSPR), Microsoft Entra Multi-factor authentication (MFA),Different Multi-factor Authentication verification methods, Azure Resources, Subscription, Management Group, Conditional Policies	2 hrs

Module 3: Navigate the Compute Services on the Microsoft Azure Platform

S.No.	Content	hrs assigned
1	Navigating to the Azure virtual machines, Data disks in Azure, Azure VMs and Interfaces, Examine the ways to create a virtual hard disk template, Creating and launching the Custom images of Azure VM, Virtual machine scale sets, Availability sets, Availability zones, Resource Locks, Resource Tags, IIS role, NSG, RBAC Roles, Azure Policies	2 hrs

Module 4: Azure Networking Services

S.No.	Content	hrs assigned
1	Overview of Azure virtual networks, Azure VNet component, Public and Private IPs, Azure VNet subnets, Manage the Azure network interface cards (NIC), Network Security Group (NSG), Routing, Managing Route tables, Service tags, Azure DNS, Manage and resolve the Private DNS	2 hrs
2	Hub and Spoke Topology, Vnet Peering, VPN- Site-to-Site, Point-to-Site, Express Route, Azure Bastion, Azure load balancers, Azure Firewall Configuration	2 hrs
3	Application gateway configuration, Application Security Groups, Azure front door service, Azure traffic manager, Network watcher	2 hrs

Module 5: Getting Started with the Azure Storage Services

S.No.	Content	hrs assigned
1	Azure Storage Account, Azure Blob Storage, Replication Strategies, Azure table storage, Azure queue storage, Azure file storage, Azure File Sync, Azure storage explorer, Azure shared access signature (SAS), Azure Databox and its use cases, Azure Import/Export service, Azure Content Delivery Network Profile with Endpoints, ARM Templates	2 hrs

Module 6: PaaS Container Services on Microsoft Azure

S.No.	Content	hrs assigned
1	Containerized web apps on Windows and Linux, App service plan, Networking for an app service and its deployment types, Deployment slots by using the Azure Portal, Azure Container Instance, Building and pushing the container images to the Azure container registry, Azure Kubernetes service, Azure Function App, Azure Logic App	2 hrs

Module 7: Working with the Azure Data Platform

S.No.	Content	hrs assigned
1	Overview of the SQL Databases, Azure SQL, Elastic Pools, Configuration, Achieve elastic scale in Azure SQL Database Cosmos DB, Multiple APIs and data models, Request units, Request Unit calculator, Partition key, Cross-partition queries, Replication and consistency levels in Microsoft Azure, Triggers, User-Defined Functions	2 hrs

Module 8: Application Management and Security

S.No.	Content	hrs assigned
1	Azure key vault, Automated Azure AD managed identities in Microsoft Entra ID, Azure AD application management	2 hrs

Module 9: Backup and Monitoring

S.No.	Content	hrs assigned
1	Recovery Services vault, Backup of Azure virtual machines, VM backup policies, Restore Azure virtual machines, Azure Monitor, Azure metric, Log Analytics workspace, Creating alerts and actions, Application Insights using the Azure Monitor	2 hrs

Module 10: Azure DevOps

S.No.	Content	hrs assigned
1	Understanding Azure DevOps concepts, Azure Boards, Azure repos, Azure pipelines, Deploying web applications using Azure pipelines.	3 hrs

AWS Solution Architect – Associate: 25 hrs

Module 1: Introduction to Cloud Architecting and Compute Service

S.No.	Content	hrs assigned
1	Introducing to Cloud Architecting , The AWS Well Architected Framework ,Best Practices for Building Solutions on AWS ,AWS Global Infrastructure , EC2 (Introduction) , Launch our First EC2 Instance with Windows, Linux, Security Group ,EBS (Introduction) ,EBS Volume and Snapshots ,AMI Types (EBS, Instance Store) ,ENI ,Encrypted Root Device Volume and Snapshot ,EC2 Pricing Model , EC2 Hibernate, CloudWatch (Introduction) , AWS Command Line (CLI)	2 hrs

Module 2: Storage Services

S.No.	Content	hrs assigned
1	S3 (Introduction) , Create an S3 Bucket ,S3 Pricing Tiers , S3 Security ,S3 Encryption , S3 Versioning , Lifecycle Management with S3, S3 Performance ,S3 Select and Glacier Sheet , AWS Organizations and Consolidated Billing , Sharing S3 Buckets Across Accounts, Cross-Region Replication, S3 Transfer Acceleration , AWS Data Sync , Creating a Static Website using Amazon S3 , EFS - Practical Lab, Amazon FSX for Windows and Amazon FSX	2 hrs

Module 3: Securing User and Application Access and Database Services

S.No.	Content	hrs assigned
1	Identity and Access Management, Create a Billing Alarm ,AWS Directory Service , IAM Policies , AWS Single Sign-on ,Advanced IAM Summary ,Databases (Introduction), configure RDS Instance ,RDS Backups, Multi-AZ	2 hrs

Module 4: Networking Services

S.No.	Content	hrs assigned
1	VPC (Introduction) ,Create your Own Custom VPC, NAT Instances ,NAT Gateways , Network Access Control Lists VS Security Groups ,Custom VPCs and ELBs ,VPC Flow Logs ,Bastions ,AWS Network Costs, Working with AWS VPC Flow Logs for Network , Creating a Basic VPC, Associated components in AWS	2 hrs
2	VPC peering same region , VPC peering another region ,VPC peering same region another account ,VPC peering another region another account ,VPN	2 hrs
3	Elastic Load Balancer, Health Checks , Advanced Load Balancer Theory, Auto Scaling Groups ,HA Architecture , Building a Fault-Tolerant WordPress Site: Setting Up EC2 , Adding Resilience and Auto Scaling , Cleaning up of resources	2 hrs
4	CloudFront (Introduction), Create a CloudFront Distribution, CloudFront Signed URLs , CloudFront Signed Cookies ,	2 hrs

	Global Accelerator ,DNS (Introduction) ,Register a Domain Name ,Route 53: Routing Policies Available on AWS , Simple Routing Policy, Weighted Routing Policy , Latency-Based Policy ,Failover Routing Policy , Geolocation Routing Policy, Geo-proximity Routing Policy (Traffic Flow only) , Multi-value Answer Policy	
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Module 5: Building Decoupled Architecture

S.No.	Content	hrs assigned
1	Simple Notification Service , Lambda Service	2 hrs

Module 6: Planning for Disaster

S.No.	Content	hrs assigned
1	Planning for Disaster, Disaster Planning Strategies ,Disaster Recovery Patterns ,Hybrid Storage and Data Migration with AWS Storage Gateway ,File Gateway	2 hrs

Module 7: Some Additional AWS Services Overview

S.No.	Content	hrs assigned
1	Code Pipeline, Code build, ECR, ECS	2 hrs
2	Systems Manager, Security hub, Config, Backup, Secrets Manager, CloudTrail, Security Hub, Certificate Manager, Billing-Costs reports	3 hrs

Module 8: Capstone Project

S.No.	Content	hrs assigned
1	Create a high availability architecture. Deploying a web server with a Dynamic Website. We will be deploying our database on Relational Database Service. After attaching the database to the web server, the client will be able to write the content on the database from the provided website.	2 hrs

GenAI with DevOps: 26 hrs**Module 1: Introduction to AI/ML**

S.No.	Content	hrs assigned
1	Overview of AI, ML, ML Algorithms- Supervised, Unsupervised , Reinforcement Learning, Deep Learning, Neural Networks	2 hrs

Module 2: Introduction to Generative AI

S.No.	Content	hrs assigned
1	Overview of Large Language Models (LLMs) (GPT, Claude, Gemini, BERT,Llama), RAG, Prompt Engineering	2 hrs
2	Diffusion Models (Stable Diffusion, DALL·E, GitHub Copilot), Use Cases in DevOps (Code Generation, Log Analysis, Incident Response, Bias, hallucinations, IP, explainability, Prompt Engineering	2 hrs

Module 3: Why Combine Gen-AI with DevOps?

S.No.	Content	hrs assigned
1	AI-Powered Automation, Self-Healing Systems, Predictive Analytics in Operations, AI-Assisted Debugging & Troubleshooting, Common Use Cases- Intelligent CI/CD assistants, Generative documentation, Code review and security scanning, AI-assisted observability (log and anomaly analysis)	2 hrs

Module 4: Gen-AI Tools & Frameworks for DevOps

S.No.	Content	hrs assigned
1	AI-Powered Code Generation & Assistance- GitHub Copilot, Amazon CodeWhisperer, AI-assisted scripting (Bash, Python, Terraform), AI for Infrastructure as Code (IaC)- Generating Terraform/Ansible scripts using AI, AI-based IaC validation & optimization with Tools- ChatGPT, Claude, Mistral for IaC, Deploy an LLM on Kubernetes (e.g., Llama 2)	2 hrs
2	AI for Log Analysis & Anomaly Detection- Anomaly Detection in Logs using GenAI (e.g., Prometheus + LLMs), Splunk AI & LogGPT, OpenTelemetry with AI-driven insights	2 hrs
3	AI in CI/CD Pipelines- Automated Test Case Generation, AI-based Security Scanning (e.g., Snyk, DeepCode), Self-Optimizing Pipelines, Using LLMs to generate/test pipeline scripts (YAML, Bash, Dockerfiles), Automating pull request reviews using AI	2 hrs

Module 5: MLOps with Generative AI

S.No.	Content	hrs assigned
1	Evolution from DevOps to MLOps, MLOps Lifecycle, Role of DevOps Engineer in MLOps and Machine Learning Pipelines	2 hrs
2	MLOps on different Cloud Platforms- AWS, Azure, GCP, Building end-to-end Machine Learning , Train your ML model in Google Colab, Managing ML Artifacts, Types of Model Artifacts in MLOps World, ML Model Metrics	2 hrs
3	Deploying with Fast API Model, ML Flow Deployment with Kubernetes, ML Flow Setup for Model Experiments, Model Monitoring, Drift of Models, Collaboration between Data Scientists and DevOps Engineers in ML Projects.	2 hrs

Module 6: Building AI Assistants for Cloud Operations

S.No.	Content	hrs assigned
1	Creating ChatOps Assistants Using GenAI and Slack/Teams Integrations and Prompts, Fine-tuning LLMs for Infrastructure Support	2 hrs
2	Use Cases- Automating Troubleshooting with Prompt-Driven Agents, Different Prompts - Automating Troubleshooting with Prompt-Driven Agents, Integrating LLMs with CI/CD Tools (Jenkins, GitLab CI)	2 hrs
3	Build a GenAI-powered ChatOps Bot	2 hrs